Tea Leaf Disease Classification Using Transfer Learning: A Comparative Study of VGG16 and ResNet50

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***Abstract*—Tea leaf diseases significantly affect agricultural productivity. This study investigates two transfer learning models, VGG16 and ResNet50, for the classification of tea leaf diseases. The dataset contains three classes, including Algal Leaf, Brown Blight, and White Spot. Experimental results indicate that VGG16 slightly performs better than ResNet50 in terms of test accuracy under limited dataset conditions. This indicates the importance of model selection for small-scale agricultural datasets.**

I. Introduction

Deep learning-based plant disease detection has become a key area in precision agriculture. Convolutional Neural Networks (CNNs), particularly pretrained architectures such as VGG16 and ResNet50, enable high classification accuracy even with limited data. This study compares both models for tea leaf disease classification.

II. Dataset and Preprocessing

The dataset consists of 368 images belonging to three classes:

- Algal Leaf(113)
- Brown Blight(113)
- White Spot(142)

All images were resized to 224×224 pixels. Data augmentations, including rotation, flipping, and color jitter, were applied. The dataset was split into 70% training, 15% validation, and 15% testing.

III. Methodology

A. Transfer Learning Models

VGG16 VGG16 pretrained on ImageNet was used. Feature extraction layers were frozen, and the final classification layer was modified for 3 classes.

ResNet50 The ResNet50 pretrained model was fine-tuned by unfreezing the last residual block and replacing the final fully connected layer.

B. Experimental Setup

The models were implemented using PyTorch with the following configuration:

- Input image size: 224×224
- Batch size: 16
- Optimizer: Adam(lr = 0.0001)
- Loss function: CrossEntropyLoss
- Training strategy: Transfer learning(VGG16 and ResNet50)
- Regularization: Data augmentation

Random seeds were fixed to ensure reproducibility of results

IV. Results and Discussion

A. Performance Comparison

TABLE I
Test Accuracy Comparison

Model	Accuracy
VGG16	85.71%
ResNet50	83.91%

The accuracy comparison presented in Table I shows that the VGG16 model achieved a test accuracy of 85.71%, slightly outperforming the ResNet50 model, which achieved 83.91%. This difference indicates that VGG16 provided better generalization performance on the tea leaf disease dataset. Since the dataset size is relatively small, the simpler architecture of VGG16 appears to be more effective than the deeper ResNet50 model, which typically requires larger datasets to fully utilize its representational capacity. These results suggest that transfer learning with moderately deep architectures can be highly effective for small-scale agricultural image classification tasks.

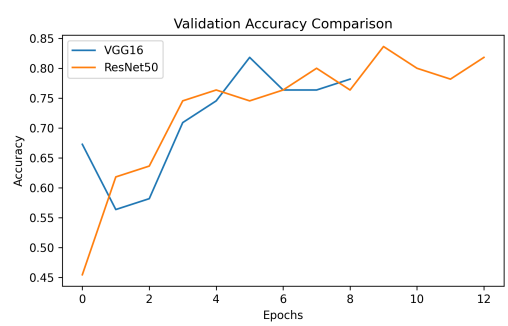

Fig. 1. Validation Accuracy Comparison of VGG16 and ResNet50

The figure illustrates the validation accuracy trends of both VGG16 and ResNet50 during training with early stopping.

Both models demonstrate effective feature learning during the initial epochs. Training was automatically terminated once the validation performance stopped improving, ensuring that the best model weights were preserved. VGG16 achieved a validation accuracy of approximately 78.18% with stable and consistent performance, while ResNet50 achieved a slightly higher validation accuracy of approximately 81.82%. The application of early stopping improves the generalization capability of ResNet50 by preventing overfitting in later epochs. However, VGG16 exhibits more stable performance throughout training, whereas ResNet50, despite higher training accuracy, still shows minor signs of overfitting.

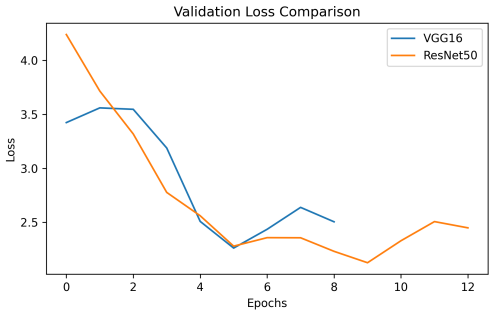

Fig. 2. Validation Loss Comparison of VGG16 and ResNet50

The figure presents the validation loss comparison between VGG16 and ResNet50 under early stopping. Both models exhibit a decreasing loss trend during the initial training stages. That indicates effective optimization. ResNet50 shows a rapid reduction in validation loss in early epochs due to its deeper architecture and residual learning capability. However, training is halted once the validation loss no longer improves, preventing further overfitting. VGG16 maintains a more stable and consistent validation loss pattern. Although ResNet50 achieves lower loss at certain stages, its slight fluctuations indicate sensitivity to the training data.

B. Confusion Matrix Analysis

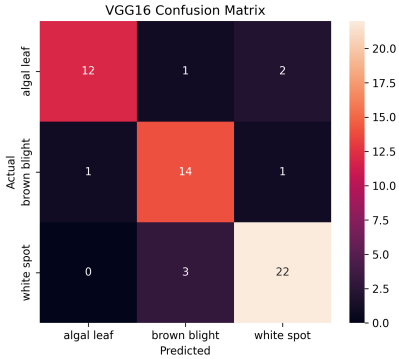

Fig. 3. Confusion Matrix of VGG16 Model

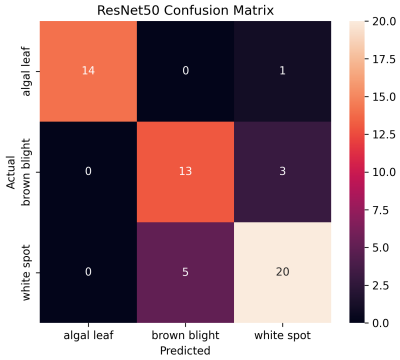

Fig. 4. Confusion Matrix of ResNet50 Model

The confusion matrices illustrate class-wise prediction performance of both models. VGG16 shows balanced classification performance across all three disease categories with fewer misclassifications compared to ResNet50. ResNet50 demonstrates strong detection capability for Brown Blight but shows reduced performance for White Spot classification. Overall, VGG16 achieves better class-level consistency, contributing to its higher overall accuracy.

V. Conclusion

This study compared VGG16 and ResNet50 for tea leaf disease classification. VGG16 achieved better performance under limited data conditions. This makes it more suitable for small agricultural datasets.

VI. Future Work

Future work may include expanding the dataset size and including advanced architectures such as EfficientNet and Vision Transformers, applying fine-tuning of deeper network layers, and deploying the trained model in mobile-based agricultural disease detection applications.

References

[1] K. He et al., "Deep Residual Learning for Image Recognition," in Proc. CVPR, 2016.

[2] GeeksforGeeks, "Introduction to Convolution Neural Network," Available: https://www.geeksforgeeks.org/machine-learning/introduction-convolution-neural-network/.

[3] Anthropic, "Claude AI Assistant," Available: https://claude.ai [Accessed: Apr. 23, 2026].